

DataAnalysis1

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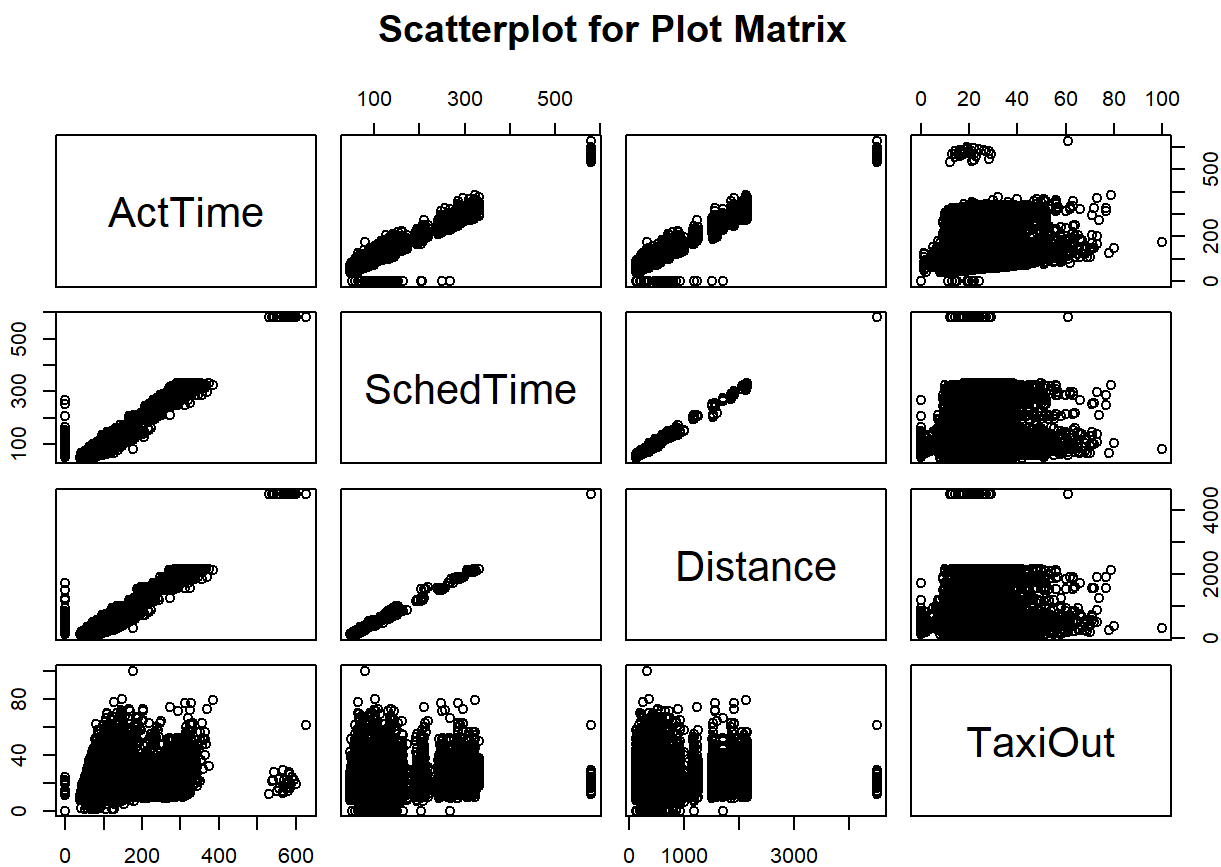
2026-02-11

Importing and Reading the Data

```
airdata <- read.csv("FullAirlineData.csv",  
                    header = TRUE,  
                    sep = ",",  
                    stringsAsFactors = TRUE)
```

Question 1: Construct a scatterplot matrix

```
plotdata <- airdata[,c("ActTime", "SchedTime", "Distance", "TaxiOut")]  
  
plotmatrix <- pairs(plotdata, main = "Scatterplot for Plot Matrix")
```



```
plotmatrix
```

```
## NULL
```

Question 2: Fit model for SchedTime vs. Distance

```
schedtimebydist <- lm(SchedTime~Distance, data = airdata)
```

Part a: State the sample the regression equation and interpret

```
schedtimebydist
```

```
##  
## Call:  
## lm(formula = SchedTime ~ Distance, data = airdata)  
##  
## Coefficients:  
## (Intercept)      Distance  
##      34.1910       0.1324
```

The coefficient for the intercept is 34.1910 and for distance it is 0.1324. Therefore the equation is

$$\hat{y} = 34.1910 + 0.1324 * Distance$$

In this context, the y-intercept of 34.1910 means that when there is 0 distance between ATL and the destination, the estimated schedule time is 34.1910. This not meaningful because having a distance of 0 is impractical to have a flight take 34.1910 minutes. In addition, it is outside the observed range of x values.

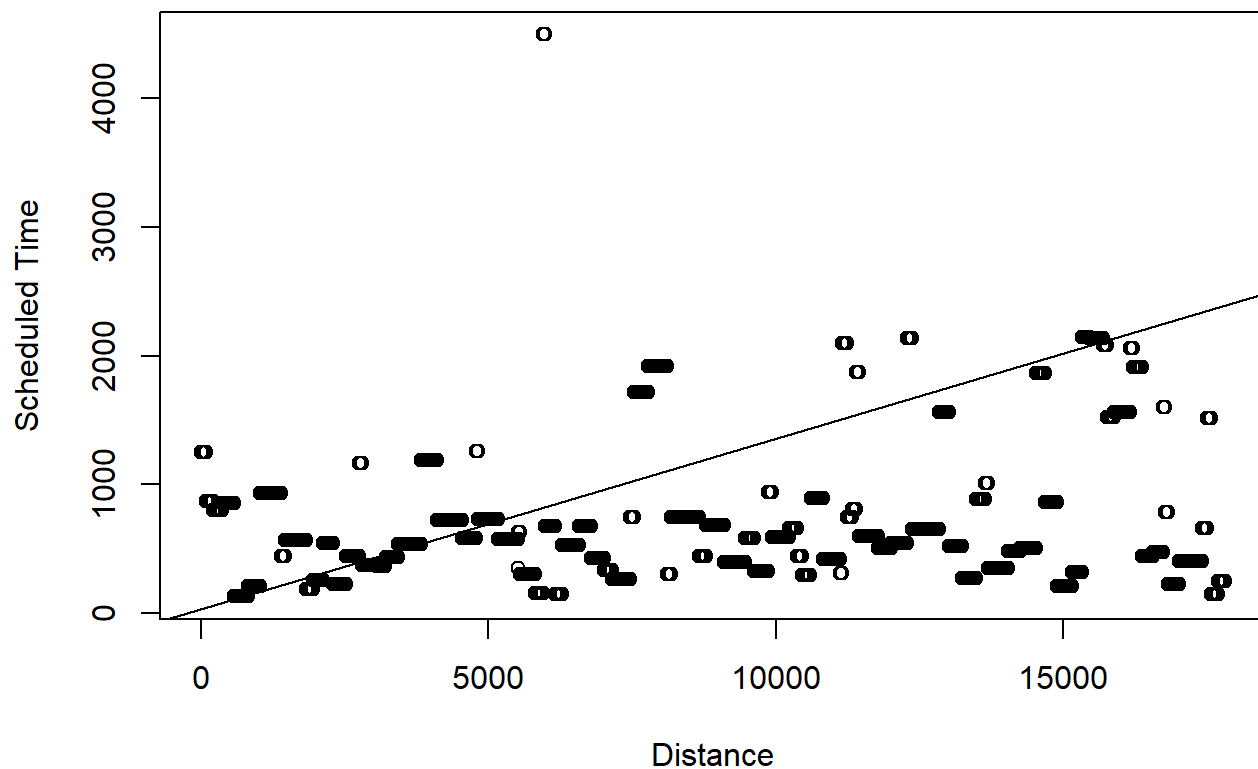
The slope coefficient is 0.1324 in this context. This means that for every 1 unit increase in distance, on average, the expected scheduled airline increases by 0.1324 minutes.

Part b: Conduct a residual analysis using assumptions

Assumption 1: Linearity Assumption

```
plot(airdata$Distance,airdata$Schedtime,main = "Scatterplot of Distance vs. Time", xlab = "Distance", ylab = "Scheduled Time")  
  
abline(schedtimebydist)
```

Scatterplot of Distance vs. Time



The first assumption needed to be checked is linearity. Based on the scatterplot, this relationship is not linear. Therefore, there is departure from the model assumption of linearity.

Assumption 2: Zero Mean Assumption

```
question1resid <- resid(schedtimebydist)
mean(question1resid)
```

```
## [1] 4.747035e-15
```

The next assumption to check is whether the mean of residuals is approximately equal to 0. The mean generated is close to zero, so this assumption is met.

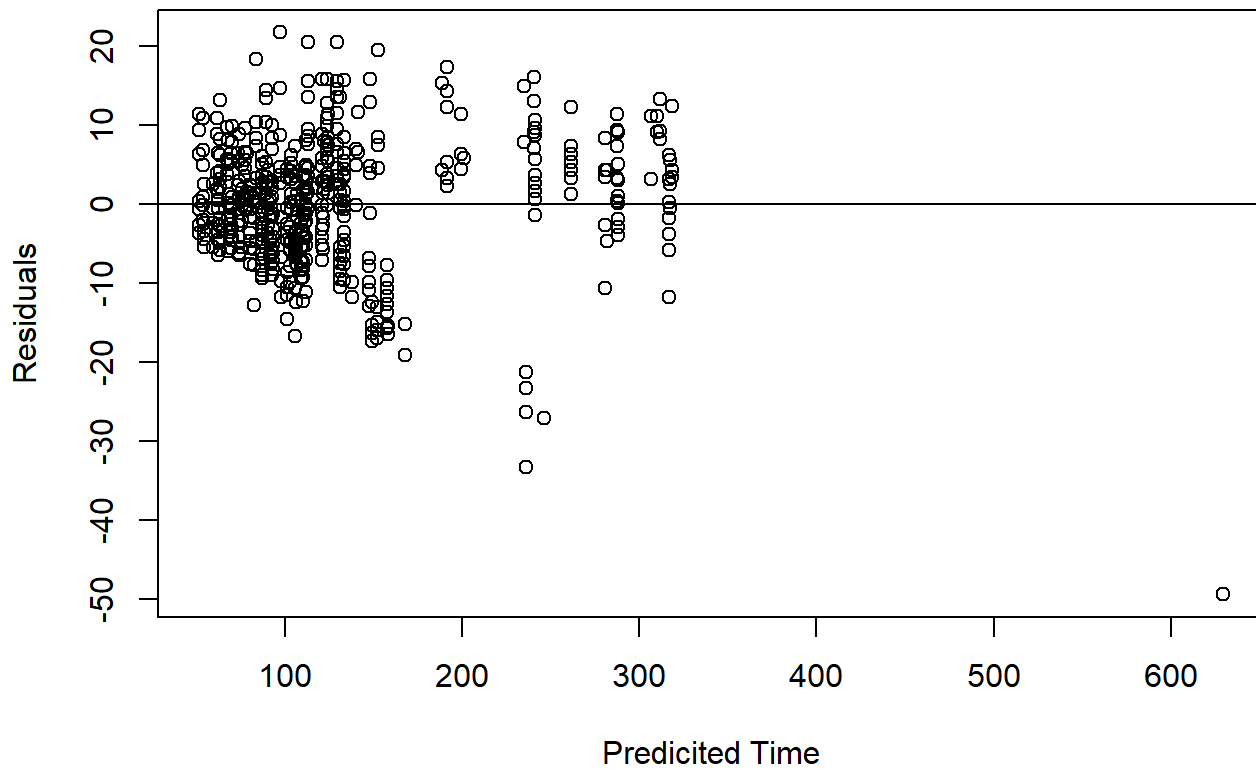
Assumption 3: Equal Variance Assumption

```
fitted1 <- schedtimebydist$fitted.values

plot(fitted1,question1resid,ylab = "Residuals", xlab = "Predicited Time", main = "Residual Plot
for Equal Variance Assumption")

abline(0,0)
```

Residual Plot for Equal Variance Assumption

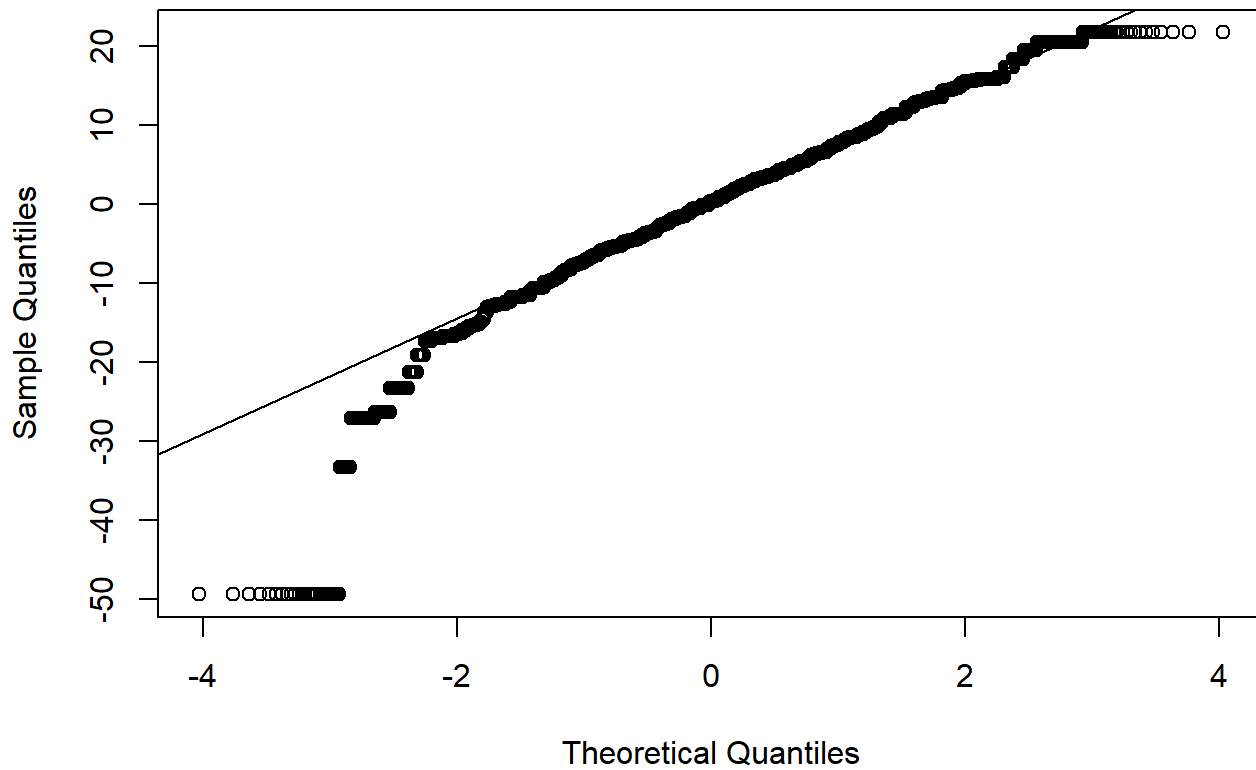


The next assumption to check is the equal variance assumption. To be met, the residual points must be equally spread around the line. From the plot, however, it can be seen that the points are not evenly spread, so there is departure from this assumption.

Assumption 4: Normality

```
qqnorm(question1resid)
qqline(question1resid)
```

Normal Q-Q Plot



The last assumption to check is whether the data points are normally distributed. From the quantile plot above, there is departure, as while many of the points follow a straight line, there are many outlier points below the line and above the line. For the most part, however, this assumption is met.

Part c: Conduct a hypothesis test

```
summary(schedtimebydist)
```

```
##
## Call:
## lm(formula = SchedTime ~ Distance, data = airdata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -49.362  -4.822   0.284   5.041  21.665
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 3.419e+01  1.011e-01   338.3  <2e-16 ***
## Distance    1.324e-01  1.145e-04  1156.3  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.974 on 17810 degrees of freedom
## Multiple R-squared:  0.9869, Adjusted R-squared:  0.9869
## F-statistic: 1.337e+06 on 1 and 17810 DF,  p-value: < 2.2e-16
```

The hypothesis test under consideration is analyzing whether the slope is meaningful in this context. To begin, the hypothesis test used is:

$$H_0 : \beta_1 = 0$$

$$H_a : \beta_1 \neq 0$$

From the summary above, the test statistic is 1156.3 and the p-value is 2e-16, which is approximately 0.00, or < 0.0001. In addition, the degrees of freedom is 17810. With all this considered, the p-value is very low, so the null hypothesis is rejected. There is enough evidence to suggest that there is a significant linear relationship between the variables.

Question 3: Fit ActTime vs. SchedTime

```
actvssched <- lm(ActTime~SchedTime, data = airdata)
```

Part a: state the sample regression coefficients and interpret

```
actvssched
```

```
##
## Call:
## lm(formula = ActTime ~ SchedTime, data = airdata)
##
## Coefficients:
## (Intercept)      SchedTime
##      0.3545         0.9895
```

The coefficient for the intercept is 0.3545 and for scheduled time it is 0.9895. Therefore the equation is $\hat{y} = 0.3545 + 0.9895 * SchedTime$

In this context, the y-intercept of 0.3545 means that when the scheduled time for a flight is 0 minutes, the estimated actual airtime of the flight is 0.3545 minutes. This not meaningful because having a scheduled flight time of 0 is impractical to have the actual flight time be 0.3545 minutes. In addition, it is outside the observed range of x values.

The slope coefficient is 0.9895 in this context. This means that for every 1 minute increase in scheduled time, on average, the expected actual airtime increases by 0.9895 minutes.

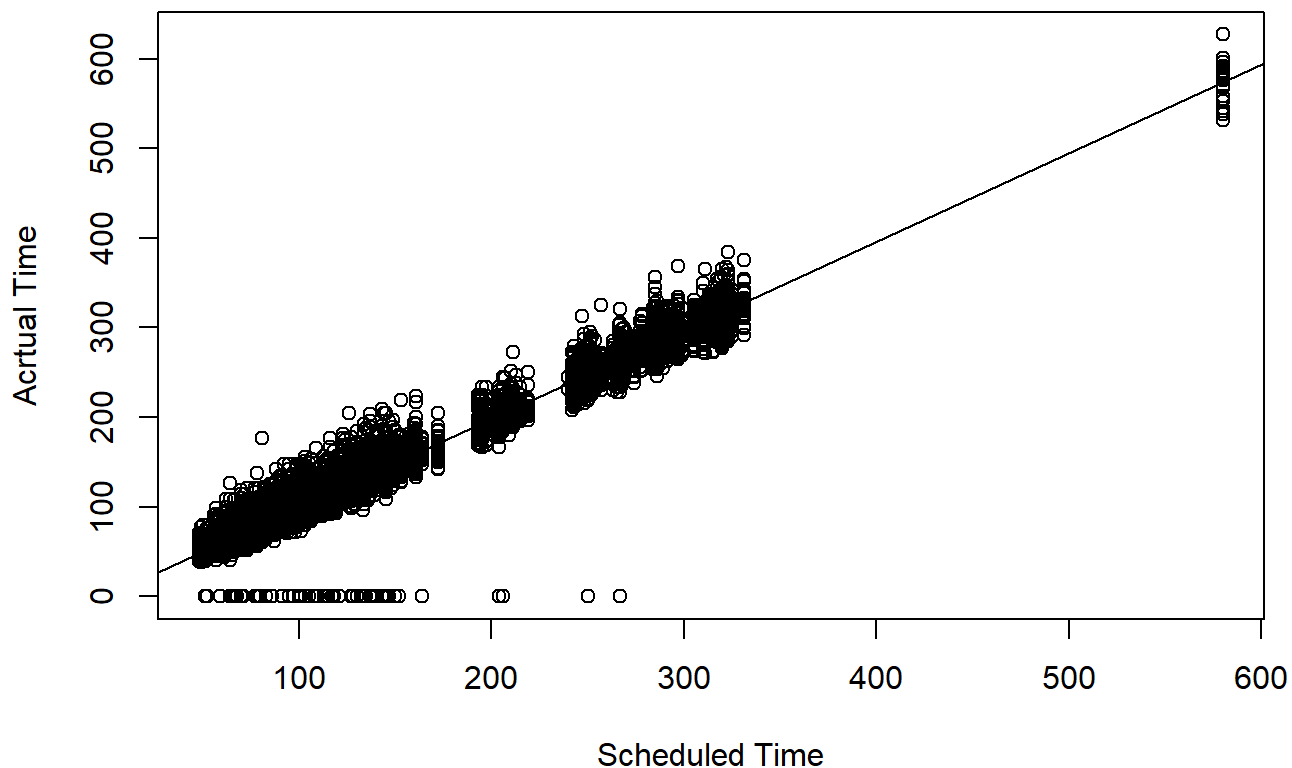
Part b: Graphical Residual Analysis

Assumption 1: Linearity Assumption

```
plot(airdata$SchedTime,airdata$ActTime,xlab = "Scheduled Time", ylab = "Acrtual Time", main = "Scatterplot of Scheduled vs. Actual Time")
```

```
abline(actvssched)
```

Scatterplot of Scheduled vs. Actual Time



The first assumption to check is whether the plot is linear. From the scatterplot, the points are approximately linear, despite some variation.

Assumption 2: Zero Mean Assumption

```
question2resid <- resid(actvssched)
```

```
mean(question2resid)
```

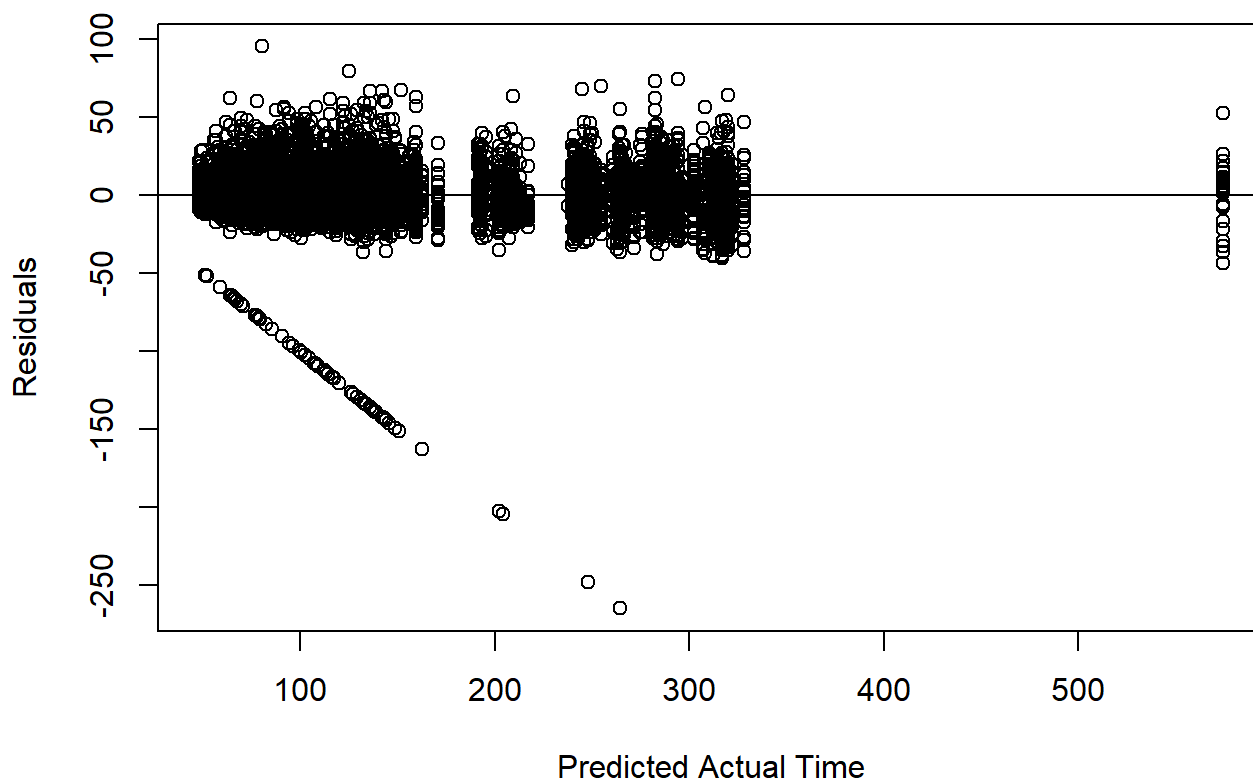
```
## [1] -8.088335e-16
```

The mean is close the zero, so this assumption is met.

Assumption 3: Equal Variance Assumption

```
fitted2 <- actvssched$fitted.values  
  
plot(fitted2,question2resid,ylab = "Residuals", xlab = "Predicted Actual Time", main = "Residual  
Plot for Equal Variance")  
  
abline(0,0)
```

Residual Plot for Equal Variance

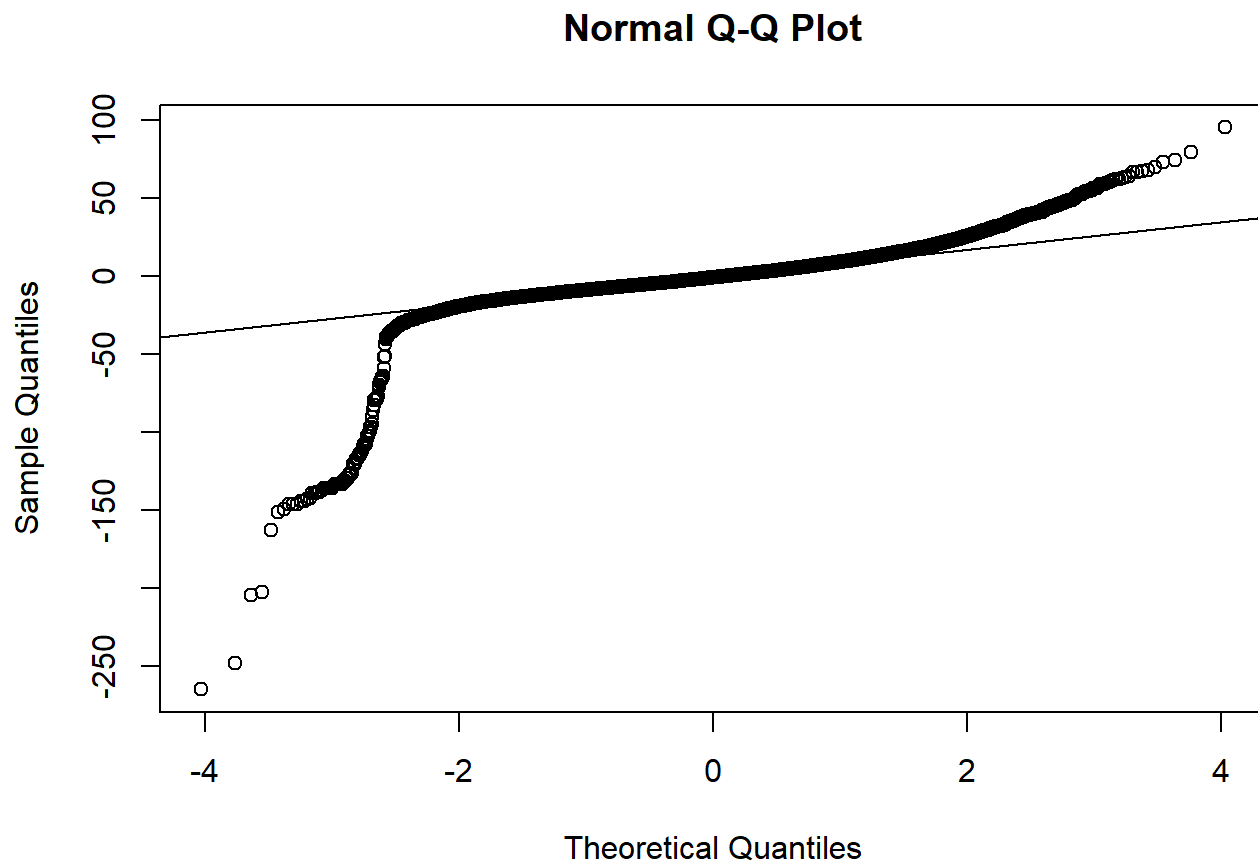


For the most part, the residuals are not consistently spread. Also, there are some points in a line that do not fit the assumption. This may be due to extreme extrapolation of outliers outside of the observed values.

The unusual diagonal of the lines in this plot is most likely the cause of when the ActTime is equal to 0. If the ActTime is equal to zero, there is more likely to be negative residuals in the plot, given the estimates will be larger. This is accounting for flights that may have been cancelled, where the actual time would be zero and the scheduled time would be more.

Assumption 4: Normality Assumption

```
qqnorm(question2resid)  
qqline(question2resid)
```



The quantile plot shows that while the majority of points are in a straight line, there are still some departures in the lower and upper tails. Therefore, this assumption is not met. There is a curve in the plot.

Question 4: Recommend a remedy and rerun regression accordingly

The remedy recommended to fix this is to rerun the data with filters and removing outliers from the data. The ActTime should have a value larger than 0. Therefore, the values should be removed from the dataset.

Part a: State the sample regression equation

```
airdatamodified <- subset(airdata, ActTime > 0)
```

```
actvssched.reg <- lm(ActTime~SchedTime, data = airdatamodified)
```

```
actvssched.reg
```

```
##  
## Call:  
## lm(formula = ActTime ~ SchedTime, data = airdatamodified)  
##  
## Coefficients:  
## (Intercept)      SchedTime  
##      0.9179      0.9896
```

The coefficient for the intercept is 0.9179 and for scheduled time it is 0.9896. Therefore the equation is
 $\hat{y} = 0.9179 + 0.9896 * SchedTime$

In this context, the y-intercept of 0.3594 means that when the scheduled time for a flight is 0 minutes, the estimated actual airtime of the flight is 0.9179 minutes. This not meaningful because having a scheduled flight time of 0 is impractical to have the actual flight time be 0.9179 minutes. In addition, it is outside the observed range of x values.

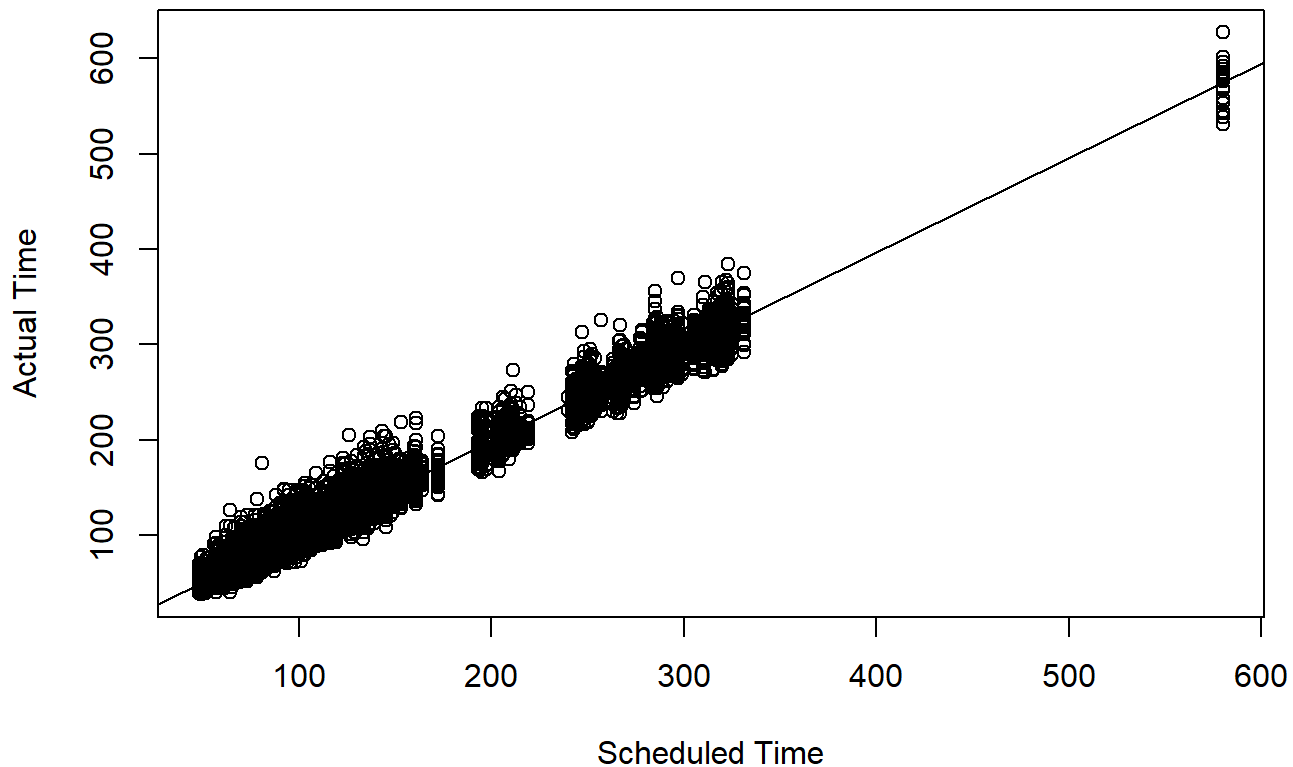
The slope coefficient is 0.9896 in this context. This means that for every 1 minute increase in scheduled time, on average, the expected actual airtime increases by 0.9896 minutes.

Part b: Graphical residual analysis

Linearity Assumption

```
plot(airdatamodified$SchedTime,  
     airdatamodified$ActTime,  
     xlab = "Scheduled Time",  
     ylab = "Actual Time",  
     main = "Scatterplot of Scheduled vs. Actual")  
  
abline(actvssched.reg)
```

Scatterplot of Scheduled vs. Actual



The scatterplot is linear, so the linearity assumption is met.

Zero Mean Error Assumption:

```
question4resid <- resid(actvssched.reg)
```

```
mean(question4resid)
```

```
## [1] 5.318523e-16
```

The mean of the residuals is $5.3e-16$, which is approximately close to 0. Therefore, this assumption is met.

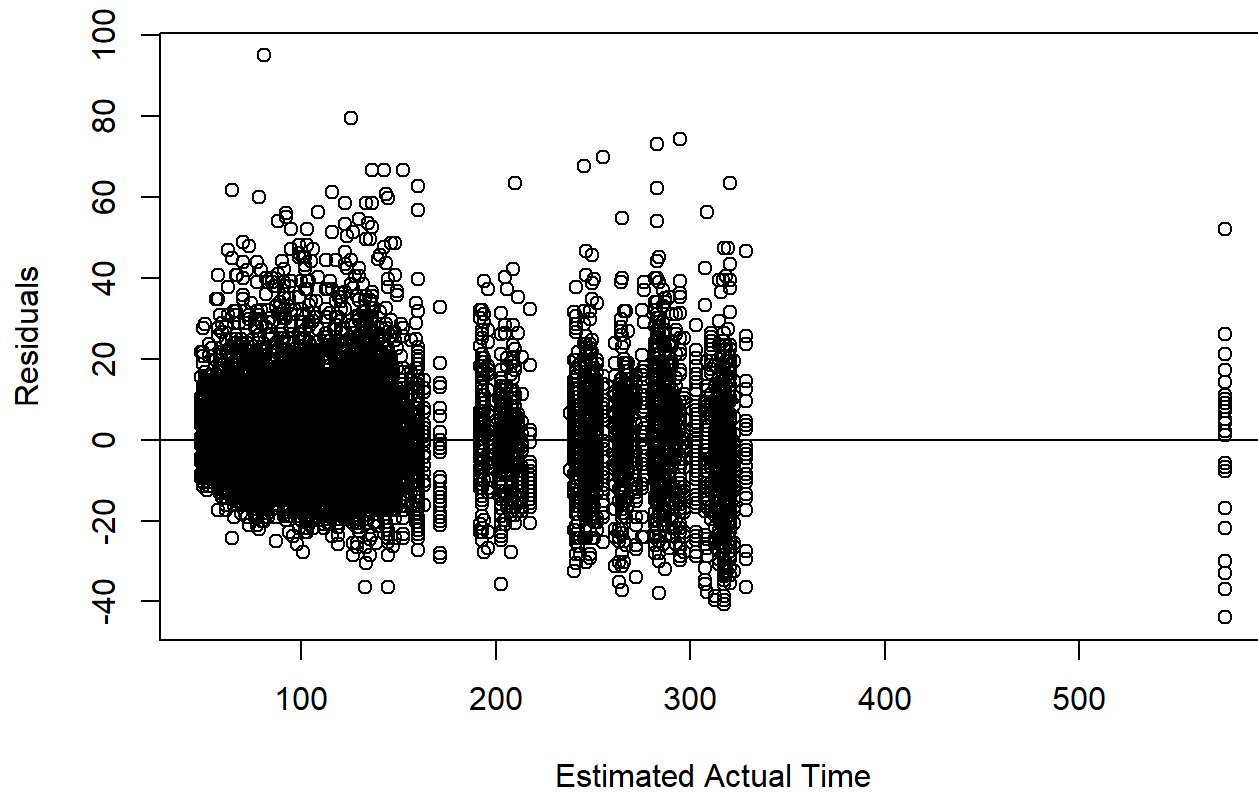
Equal Variance Assumption:

```
question4fitted <- actvssched.reg$fitted.values
```

```
plot(question4fitted,question4resid, xlab = "Estimated Actual Time", ylab = "Residuals", main =  
"Equal Variance Plot")
```

```
abline(0,0)
```

Equal Variance Plot

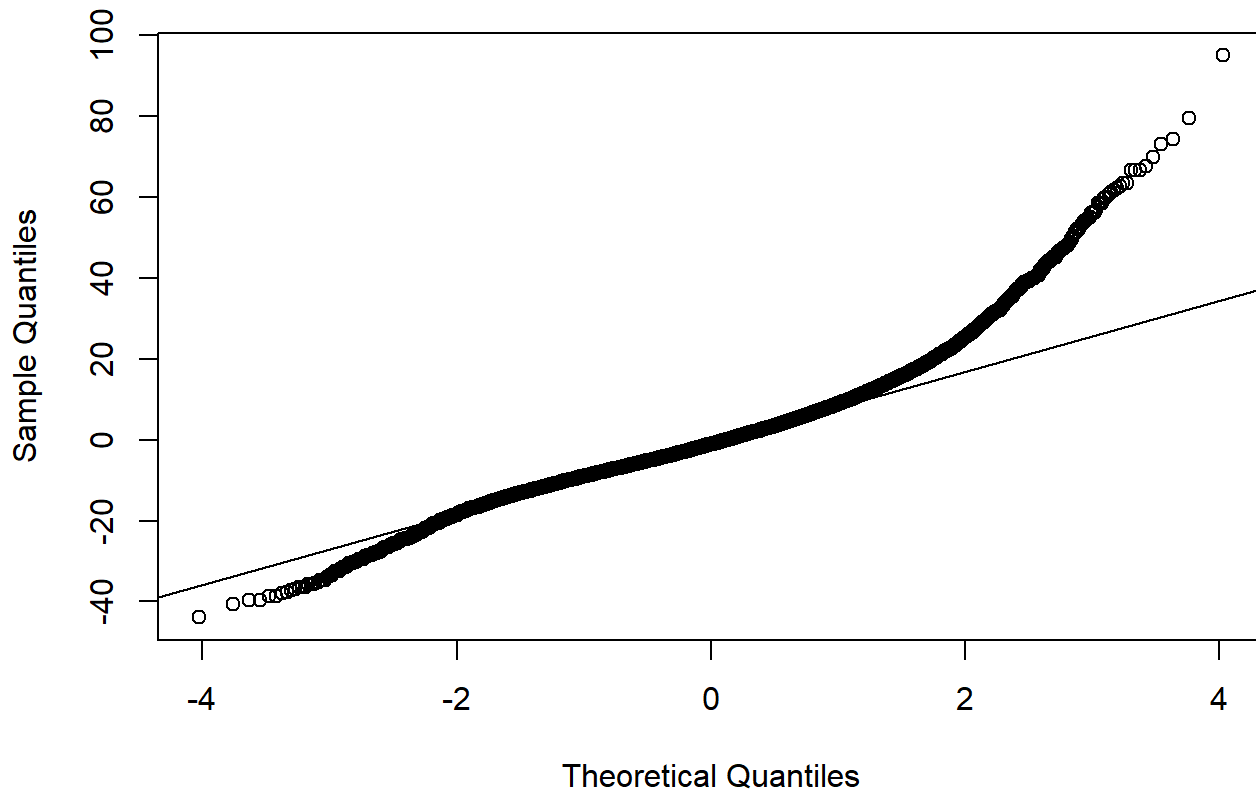


The equal variance assumption is not met. The spread is not consistent across the line.

Normality Assumption:

```
qqnorm(question4resid)  
qqline(question4resid)
```

Normal Q-Q Plot



This plot shows a more normal pattern than the previous one, however, there is still some deviation from a straight line in the upper tail. Therefore, this is departure from this assumption with the curve.

Question 5: New Variables

```
airdatamodified$ln_ActTime <- log(airdatamodified$ActTime)

airdatamodified$ln_SchedTime <- log(airdatamodified$SchedTime)
```

Part a: State the sample regression equation and interpretation

```
log.reg <- lm(ln_ActTime~ln_SchedTime, data = airdatamodified)

log.reg
```

```
##  
## Call:  
## lm(formula = ln_ActTime ~ ln_SchedTime, data = airdatamodified)  
##  
## Coefficients:  
## (Intercept) ln_SchedTime  
##      0.00673      0.99729
```

The coefficient for the intercept is 0.00673 and for log of scheduled time it is 0.99729. Therefore the equation is $\log \hat{y} = 0.00673 + 0.99729 * \ln \text{ScheduledTime}$

In this context, the y-intercept of 0.00673 means that the scale of the actual and scheduled times are very close to each other. Actual time, on average is estimated to be approximately 0.67% larger than scheduled time.

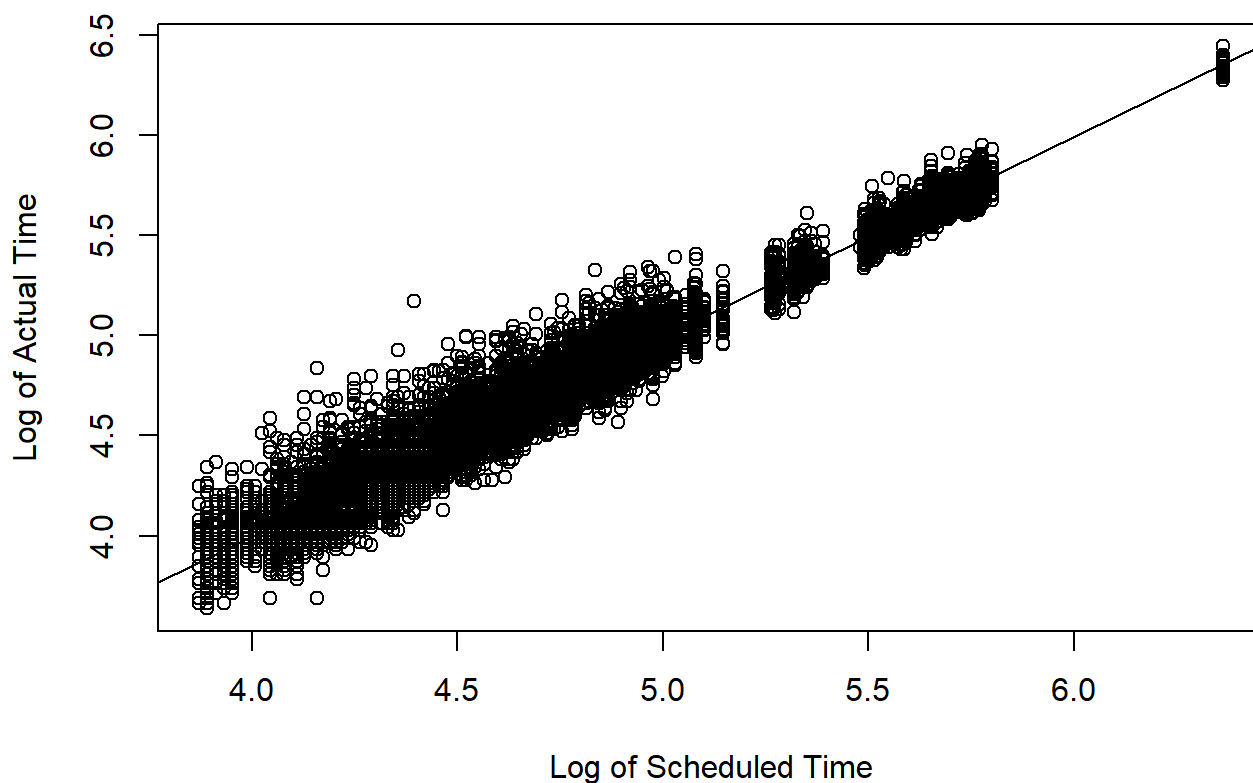
The slope coefficient is 0.99729 in this context. This means that a 1% increase in scheduled time results in an expected actual time increase of 0.997%, on average.

Part 2: Graphical Residual Analysis

Linearity Assumption:

```
plot(airdatamodified$ln_SchedTime, airdatamodified$ln_ActTime, xlab = "Log of Scheduled Time", ylab = "Log of Actual Time", main = "Scatterplot of Logs")  
  
abline(log.reg)
```

Scatterplot of Logs



The linearity assumption is met, since the plot is approximately linear.

Zero Mean Error Assumption:

```
log.res <- resid(log.reg)

mean(log.res)
```

```
## [1] -1.792018e-17
```

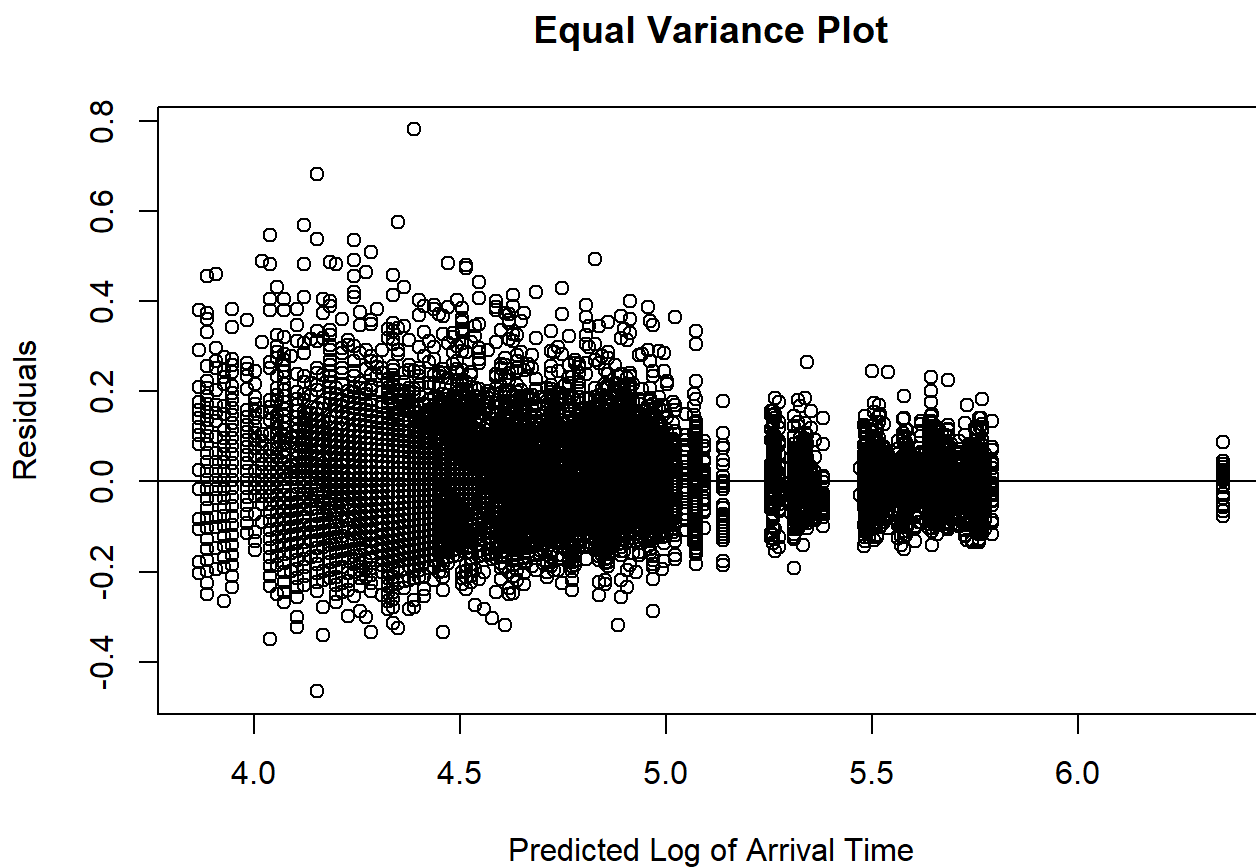
The mean of residuals is approximately -1.78×10^{-17} , which is close to 0. Therefore, this assumption is met.

Equal Variance Assumption

```
logfitted <- log.reg$fitted.values

plot(logfitted,log.res,xlab="Predicted Log of Arrival Time",ylab = "Residuals", main = "Equal Va
riance Plot")

abline(0,0)
```

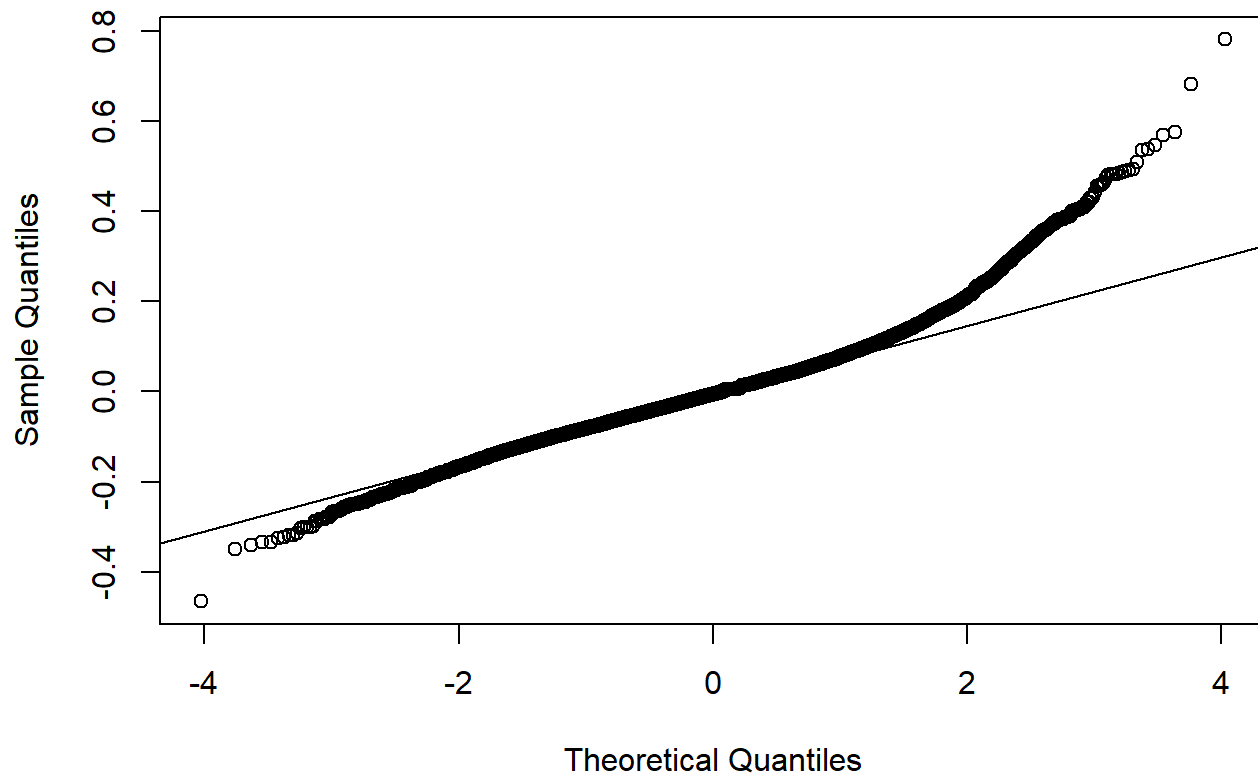


The points take on a bit of a funnel shape, so this assumption is not met.

Normality Assumption:

```
qqnorm(log.res)  
qqline(log.res)
```

Normal Q-Q Plot



There is departure from normality, since there is curvature in the plot.